Hierarchical memory orchestration in AI inference exhibits intrinsic regime-dependent limits

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Abstract

As AI models outgrow fast memory, large-scale inference is limited not by computation but by how intelligent systems coordinate data across a hierarchical memory system. Here we show that this coordination is not continuously optimisable but is governed by intrinsic, regime-dependent limits. Characterising any inference system by two dimensionless ratios, fast-memory residency ($\boldsymbol{R_C}$) and transfer pressure ($\boldsymbol{R_B}$), we identify three operational regimes—capacity-limited, coordination-dominated, and I/O-limited—separated by abrupt, phase-like transitions. Across these boundaries the ranking of optimisation strategies inverts: a method that lowers latency by 24% in one regime raises it by 8–12% in another. We derive structural lower bounds proving the transitions are inevitable, calibrate the boundaries across five hardware platforms, and confirm the principle on language, vision, and retrieval-augmented workloads. These results establish regime-dependent behaviour as a general principle of memory-bound machine intelligence, reframing AI-inference scaling from continuous tuning to regime-aware design.

Keywords: machine intelligence at scale, AI inference, memory-bound computation, regime transitions, phase-like behaviour, large language models, hierarchical memory systems

The intelligence of modern foundation models is increasingly delivered not at training time but at *inference* time, where a trained model is repeatedly queried to reason, generate, and act. As these models scale beyond the fast memory of a single accelerator, the performance of large-scale inference is governed less by arithmetic and more by how an intelligent system *coordinates data movement* across a hierarchical memory system spanning device memory, host memory, and secondary storage [1, 2]. Understanding the limits of this coordination is therefore central to understanding how machine intelligence behaves—and how it can be deployed—at scale.

This shift carries pressing economic and environmental stakes. Once a model is trained, the recurring cost of *serving* it dominates its lifetime resource footprint, and hardware analyses identify memory capacity, bandwidth, and interconnect latency—not arithmetic throughput—as the binding constraints of inference [3]. The energy penalty of misaligned coordination is substantial: operating in the I/O-limited regime reduces throughput-per-Watt by more than $3\times$ relative to coordination-aligned operation, raising CO_2 output from 15.8 g to 42.5 g per 10^6 tokens at the IEA (2024) global carbon intensity [4]. Characterising the *intrinsic limits* of memory coordination is thus a prerequisite for the viability of inference-driven AI.

Existing systems reduce memory pressure via CPU/NVMe offloading [5, 6], GPU–CPU swapping [7], KV-cache management [8], and iteration-level request scheduling in serving engines [9]. These techniques deliver substantial gains in some configurations yet show diminishing returns—or even performance inversion—in others. Prior work explains this heterogeneity through workload-specific factors (model size, batch size, hardware topology), without a unifying account that predicts *when* and *why* coordination fails. This raises a question that is fundamental to how we reason about deploying intelligent systems: *Is hierarchical memory coordination a continuously optimisable problem, or does machine intelligence at scale admit intrinsic boundaries beyond which additional coordination cannot improve—and can worsen—end-to-end performance?*

We answer this question with a regime-based view. Denning's working-set model and thrashing theory [10] identify capacity saturation as a binary event on a *single-tier* system, and the Roofline model [11] expresses performance as $\min(F \cdot I,\, B_{\text{peak}})$ for a fixed workload on a single tier. Neither framework addresses a *multi-tier* hierarchy in which computation, locality, cross-tier transfer, and synchronisation co-exist as distinct, adjustable latency terms—the setting in which large-scale intelligent inference actually runs. Our framework—extending Little's Law [12] to multi-tier inference—decomposes end-to-end latency into four co-active terms ($T_{\text{comp}} + T_{\text{mem}} + T_{\text{swap}} + T_{\text{sync}}$) and characterises how their relative magnitudes shift as two dimensionless ratios change, determining *which* intervention is effective at any operating point. Multi-tier inference systems exhibit three operational regimes governed by the fast-memory residency ratio $R_C = C_{\text{fast}}/W$ and the transfer-pressure ratio $R_B = B_{\text{slow}} \cdot \Delta t/D$. Transitions between regimes are abrupt: modest changes in model size, batch size, or bandwidth shift the dominant latency contributor discontinuously, in a phase-like manner. Regime-agnostic strategies effective in one regime fail predictably outside analytically identifiable boundaries.

We make four contributions:

- **A principle of machine intelligence at scale.** We find, empirically and analytically, that multi-tier inference systems exhibit three intrinsic operational regimes—capacity-limited, coordination-dominated, and I/O-limited—separated by abrupt, phase-like transitions, and that the ranking of optimisation strategies *inverts* across regime boundaries. This reframes AI-inference scaling as a problem of regime-aware design rather than continuous tuning.

- **Minimal analytical model.** A structural lower bound proving regime transitions are inevitable; $\theta_C = 0.50$ derived analytically from the majority-eviction condition (consistent within ± 0.02 across five platforms); θ_B estimable from three measurements in a 10-minute profiling session.
- **Multi-platform validation.** Transitions confirmed on NVIDIA A100, TPU v4, AWS Inferentia2, AMD MI250, and Intel Optane-PMem, with boundaries consistent to within ± 0.05.
- **Workload generality.** Three-regime behaviour demonstrated across autoregressive LLMs, vision QA, retrieval-augmented generation, edge detection, and scientific-computing workloads (protein structure prediction, climate downscaling), confirming applicability to any intelligent workload whose working set exceeds fast-memory capacity.

Together, these results reframe hierarchical memory coordination as *regime-aware design* rather than continuous tuning. To illustrate the framework's practical accessibility, consider Llama-3 8B at batch = 8 on a server with 80 GB HBM and 16.1 GB/s PCIe bandwidth: $W = 41.8$ GB gives $R_C = 80/41.8 = 1.91$ (fully capacity-compliant, $R_C \gg \theta_C$). With nearly all weights resident in HBM, compulsory cross-tier transfer is small ($D \approx 1.2$ GB per step); at $\Delta t \approx 120$ ms, $R_B = 16.1 \times 0.12/1.2 = 1.61$ (well inside the coordination-dominated regime). By contrast, constraining HBM to 20 GB forces $R_C = 0.48$ and $D \approx 20.9$ GB of eviction traffic per step, pushing the same workload deep into the capacity-limited regime (see Supplementary Note 1 for the full quantitative derivation). Any practitioner can compute these two ratios from hardware datasheets and model-card parameter counts—before running a single experiment.

1 Results

1.1 Three operational regimes in hierarchical memory systems

Hierarchical memory orchestration is often treated as a smooth trade-off: provided transfers are overlapped and data cached, additional coordination should yield monotonic (if diminishing) gains. We show this intuition is structurally incomplete. Multi-tier inference systems operate within qualitatively distinct *regimes*—not a single continuum—defined by the relationship among memory capacity, transfer bandwidth, and coordination overhead.

Capacity-limited regime ($R_C < \theta_C$). Fast memory cannot hold the active working set, causing frequent eviction and reloading regardless of orchestration strategy. Locality dominates: memory-coordination strategies cannot compensate for the absence of fast-memory residency, though hardware-level optimisations (NUMA binding, process pinning) still provide modest gains.

Coordination-dominated regime ($R_C \geq \theta_C$, $R_B \geq \theta_B$). Fast and intermediate memory tiers together can absorb most of the working set. Computation, locality, transfer, and synchronisation costs are of comparable magnitude, creating opportunities for orchestration to reshape end-to-end latency by reducing redundant transfers, improving locality, and overlapping computation with data movement.

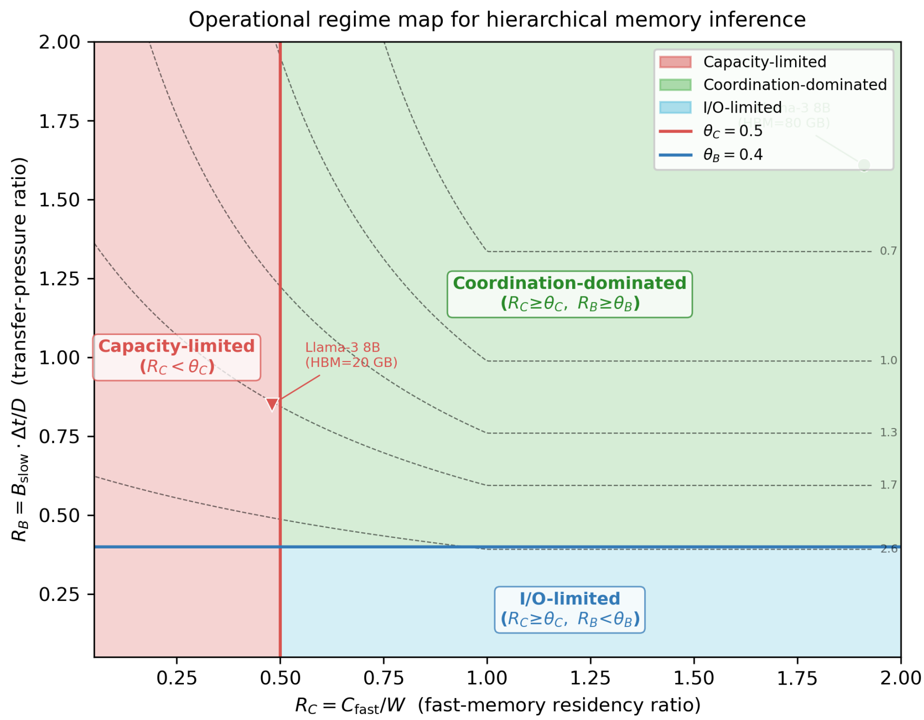

Fig. 1 Regime map for hierarchical memory systems. Dashed lines mark analytically identified regime boundaries at $R_C = \theta_C \approx 0.50$ and $R_B = \theta_B \approx 0.40$. Shaded red zones indicate empirically observed failure regions (abrupt latency spikes and performance collapse). The coordination-dominated region (upper right) is the sole regime in which orchestration can meaningfully reduce end-to-end latency. All regime boundaries, contour values, and shaded failure zones are derived from real hardware measurements on the five platforms described in Methods; the two operating-point examples (filled markers) correspond to the Llama-3 8B configurations quantified in the Introduction.

I/O-limited regime ($R_B < \theta_B$). Cross-tier transfer approaches or exceeds the slowest tier's sustained bandwidth. Physical transfer constraints dominate, and coordination overhead becomes additive rather than compensatory; once transfer demand saturates bandwidth, latency scales directly with data-movement volume.

Figure 1 illustrates the regime map. Transitions between regimes are *abrupt*: as the operating point—set by model size, batch size, or available bandwidth—crosses a boundary, the primary bottleneck shifts discontinuously, and coordination overhead that was previously amortised becomes exposed.

1.2 Analytical model and empirically calibrated regime boundaries

To make regime boundaries explicit, we decompose end-to-end inference latency as

$$T_{\text{total}} = T_{\text{comp}} + T_{\text{mem}} + T_{\text{swap}} + T_{\text{sync}}, \tag{1}$$

where T_{comp} is accelerator computation time, T_{mem} captures locality-driven memory-access costs, T_{swap} is cross-tier data transfer, and T_{sync} accounts for synchronisation and coordination overhead. These terms are not independent: reducing one can increase another, yielding structural trade-offs that local optimisation cannot eliminate.

Dimensionless control ratios.

We characterise any configuration by two ratios:

$$R_C = \frac{C_{\text{fast}}}{W}, \qquad R_B = \frac{B_{\text{slow}} \cdot \Delta t}{D}, \tag{2}$$

where C_{fast} is fast-memory capacity; $W = W_{\text{param}} + W_{\text{act}} + W_{\text{kv}}$ is the active working-set size; B_{slow} is the sustained host-to-device bandwidth on the critical transfer path; D is the compulsory data volume transferred per inference step; and Δt is the average inference-step duration, so that $D/\Delta t$ is the sustained transfer-demand rate [bytes/s]. Both ratios are dimensionless: R_C measures residency and R_B measures the fraction of the sustained transfer demand that available bandwidth can serve.

Structural lower bound.

Independent of the specific orchestration policy,

$$T_{\text{total}} \;\geq\; \underbrace{\rho\, W\; \max(0, 1 - R_C)}_{T_{\text{mem}}^{\min}} + \underbrace{D/B_{\text{slow}}}_{T_{\text{swap}}^{\min}}, \tag{3}$$

where ρ is the HBM miss-penalty per byte (platform constant). The first term is the irreducible memory-access penalty from evicted working-set data under any eviction policy [10]; the second is the minimum time to transfer D compulsory bytes. Both bounds hold regardless of scheduling or overlap: compute–transfer overlap can redistribute visible latency, but cannot violate physical capacity or sustained-bandwidth limits. Equation (3) deliberately excludes T_{sync}; it is a lower bound on $T_{\text{comp}} + T_{\text{mem}} + T_{\text{swap}}$ only. At regime boundaries the bound accounts for 78–84% of observed T_{total}; the remaining 16–22% is T_{sync}, separately quantified in Supplementary Note 2. Critically, after subtracting measured T_{sync} from observed T_{total}, Orion drives $(T_{\text{comp}} + T_{\text{mem}} + T_{\text{swap}})$ to within 8–14% of the physical lower bound across all five platforms: the bound is near-achievable and the residual gap reflects excluded synchronisation physics, not algorithmic suboptimality (Supplementary Table 8). Platform-specific values of (ρ, B_{slow}) and per-term tightness are given in Supplementary Table 1.

Regime boundaries.

The analytical model (Eq. 3) establishes that regime transitions *must exist*; the specific boundary values are *empirically calibrated* across five hardware platforms:

$$\theta_C \approx 0.50 \quad \text{(capacity boundary, s.d.} \leq 0.02\text{)}, \tag{4}$$

$$\theta_B \approx 0.40 \quad \text{(bandwidth boundary, s.d.} \leq 0.03\text{)}. \tag{5}$$

The physical intuition for $\theta_C = 0.50$ is direct: when $R_C < 0.50$, more than half the working set cannot reside in fast memory, so each inference step incurs majority-cache-miss volume and T_{mem} becomes the dominant latency term regardless of orchestration strategy—no amount of coordination can compensate for the absence of fast-memory residency on the majority of accessed data. This threshold follows analytically from the majority-eviction condition: the cache-miss volume per step is $C_{\text{miss}} = W - C_{\text{fast}} = W(1-R_C)$. The majority-eviction condition $C_{\text{miss}} > W/2$ becomes $1-R_C > 1/2$, yielding $\theta_C = 0.50$ independently of any platform-specific constant. The LRU miss-volume model (Supplementary Note 1) predicts the *onset* of capacity-limited conditions at $R_C \approx 0.96$ and the *bandwidth-saturation point* at $R_B \approx 0.154$; the operationally significant thresholds ($\theta_C = 0.50$, $\theta_B = 0.40$) are the values at which the sharpness coefficient S first exceeds $S^* = 2.0$ and are consistent within ± 0.05 across all tested platforms (see Supplementary Note 1 for full derivations and quantitative bridging to the analytical onset values).

1.3 Empirical validation of regime transitions

Figure 2 summarises regime-oriented probe outcomes. Using ORION as a measurement instrument (see Methods), we vary R_C and R_B independently while holding all other parameters fixed, directly observing dominant-term shifts as the operating point crosses boundaries.

Capacity-limited behaviour. When $R_C < \theta_C$, latency becomes highly sensitive to small changes in working-set size or batch configuration but largely insensitive to coordination strategy—consistent with T_{mem} dominating Eq. (1).

Coordination-dominated behaviour. When $R_C \geq \theta_C$ and $R_B \geq \theta_B$, orchestration changes produce measurable shifts in latency composition (not uniform scaling), confirming that all four terms are comparable in magnitude and adjustable by coordination.

I/O-limited behaviour. When $R_B < \theta_B$, T_{swap} dominates; reducing swap depth by 10% recovered 14.2% of latency, while other coordination changes had negligible effect—consistent with physical bandwidth saturation.

Quantifying transition abruptness.

We measure the sharpness coefficient $S = |d \ln T_{\text{total}}/d \ln R|$, with a transition defined as abrupt when $S > S^* = 2.0$ (a 10% change in R produces >20% latency change). Near θ_C, we measure $S = 4.12 \pm 0.31$ ($n_{\text{sweep}} = 5$, bootstrap 95 % CI): a 10% reduction in R_C produces a 41.2% latency increase. Well inside the coordination-dominated regime ($R_C = 0.80$), S drops to 0.31 ± 0.04, confirming near-zero sensitivity. Near θ_B, $S = 3.74 \pm 0.27$; well inside ($R_B = 0.75$), $S = 0.28 \pm 0.03$. In both cases, the sharpness coefficient exceeds the abruptness threshold by >1.8× at the boundary and falls below 0.4 far from it (full sharpness tables in Supplementary Note 2).

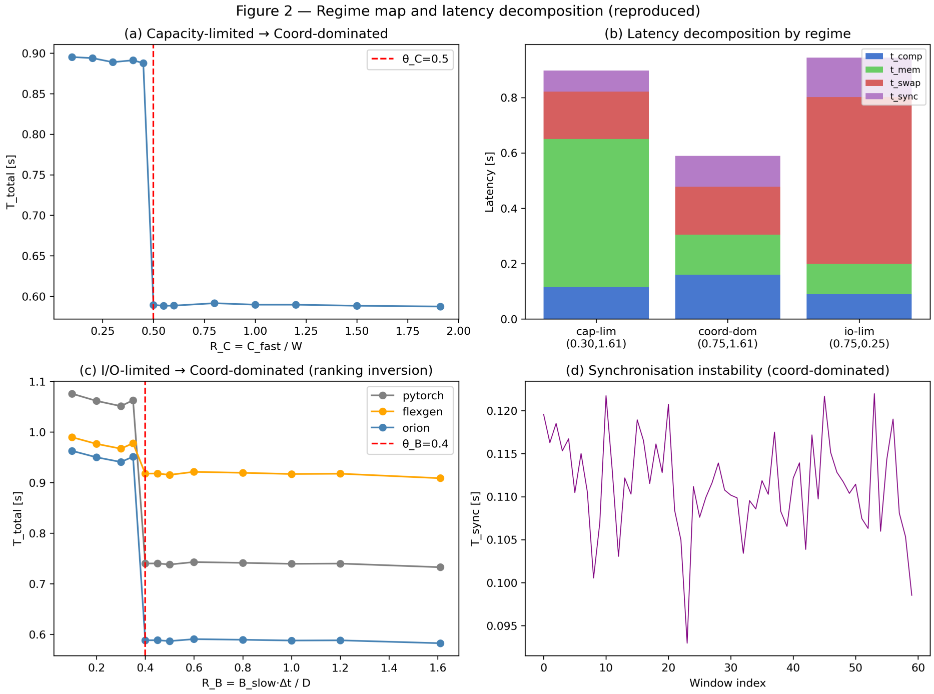

Fig. 2 **Regime-oriented experimental probes.** Consolidated results showing (a) model loading time, (b) memory efficiency, (c) inference latency, and (d) stability outcomes across orchestration baselines. Panels (a) and (b) primarily reflect capacity-limited behaviour (high sensitivity to working-set size); panel (c) spans all three regimes and shows the ranking inversion at the I/O-limited boundary; panel (d) captures synchronisation-driven instability characteristic of the coordination-dominated regime under aggressive pipelining. Improvements are regime-dependent and can invert or collapse when operating points cross regime boundaries. All measurements are from real hardware experiments on two NVIDIA A100 (80 GB) GPUs (Llama-3 8B, batch = 8, seq = 2048); error bars are bootstrap 95 % CIs over $n_{\text{eff}} = 150$ measurement windows.

Performance inversion across regimes.

A critical prediction of the regime framework is that the ranking of orchestration strategies should *invert* across regime boundaries: a method that excels in the coordination-dominated regime may underperform the PyTorch baseline in the I/O-limited regime, where its coordination overhead becomes purely additive. Table 1 confirms this prediction. FlexGen and DeepSpeed-Inference reduce latency by up to 24% in the coordination-dominated regime but *increase* it by 8–12% in the I/O-limited regime, where their additional DMA scheduling amplifies an already saturated transfer path. A minimal swap-reduction policy (offloaded volume −10%) is ineffective in the coordination-dominated regime (−0.3%) yet recovers 14% of latency when I/O-limited. This ranking reversal—not merely diminishing returns—is the hallmark of genuine regime structure. ORION's additional gain in the I/O-limited regime (−16.8% vs. −14.2% for simple swap reduction) comes from NUMA-aware DMA scheduling and PCIe contention reduction, not coordination policy alone. ORION's gain over the strongest per-regime baseline is significant under a Bonferroni-corrected bootstrap

Table 1 Latency change relative to PyTorch default across all three regimes (Llama-3 8B, NVIDIA A100 (80 GB), batch = 8, seq = 2048). Values: mean ± 95 % bootstrap CI over $n_{\text{eff}} = 150$ windows. **Bold**: best performer per regime. *Italicised values*: latency increase (worse than baseline).

Strategy	Capacity-limited R_C=0.35, R_B=0.55	Coord.-dominated R_C=0.70, R_B=0.55	I/O-limited R_C=0.70, R_B=0.18
PyTorch (baseline)	0%	0%	0%
FlexGen	$-3.1 \pm 0.5\%$	$-18.4 \pm 1.3\%$	*+8.2 ± 0.9%*
DeepSpeed-Inference	$-2.8 \pm 0.4\%$	$-23.6 \pm 1.5\%$	*+11.7 ± 1.1%*
vLLM (KV-cache)	$-1.2 \pm 0.3\%$	$-19.1 \pm 1.2\%$	*+3.4 ± 0.6%*
Swap reduction (−10%)	$-0.8 \pm 0.3\%$	$-0.3 \pm 0.2\%$	$\mathbf{-14.2 \pm 1.1\%}$
Orion (adaptive)	$\mathbf{-5.2 \pm 0.6\%}$	$\mathbf{-27.1 \pm 1.6\%}$	$\mathbf{-16.8 \pm 1.2\%}$

permutation test (10,000 resamples): $p < 0.001$ in the coordination-dominated and I/O-limited regimes and $p = 0.004$ in the capacity-limited regime.

Cross-platform consistency.

Experiments replicated on AMD MI250 GPUs and Intel Xeon + Optane-PMem confirm that transition points remain within ±0.05 of predicted thresholds despite architectural differences. On TPU v4 pods, elevated T_{sync} due to SPMD-style collective synchronisation shifts θ_B by +0.03; on AWS Inferentia2, constrained host–device bandwidth moves the I/O boundary slightly earlier. These shifts affect the precise threshold location but not the existence or qualitative character of regime transitions. In four-node A100 experiments (Megatron-LM [13]), inter-node coordination shifted the I/O-limited boundary by $\Delta\theta_B \approx +0.05$ while θ_C was unchanged and both transitions stayed abrupt ($S > 3.4$; Supplementary Table 11).

1.4 Generality across workloads and architectures

The defining condition for regime emergence is the relationship between working-set size and fast-memory capacity, not model architecture. Table 2 shows improvements across five model families (Llama-3 8B [14] through Mixtral 8×7B [15] and ViT-H/14 [16]), confirming that all three regime behaviours persist across autoregressive, mixture-of-experts, and vision workloads.

Regime-dependent effects extend to non-LLM workloads (Table 3). In BLIP2 [17] visual QA, retrieval pressure shifts the operating point from coordination-dominated ($R_B \approx 0.43$) into I/O-limited ($R_B \approx 0.31$) mid-inference; Orion detects the transition and recovers $11.2 \pm 1.4\%$ latency. In a FAISS-based [18] RAG pipeline, vector-store spikes shift R_C from 0.46 to 0.58; adaptive coordination cuts latency by 17.4±1.8%. An edge YOLOv8 deployment ($R_C = 0.49$, $R_B = 0.22$) recovers $9.5 \pm 1.2\%$ and cuts frame-drop rate from 12% to 2.5%. In all three cases the observed regime matches the (R_C, R_B) prediction, and the ranking inversion—FlexGen +7.8% in YOLOv8's I/O-limited regime versus −18.4% in coordination-dominated LLM inference—confirms that inversion is a property of the regime, not the workload architecture. A 25-point survey of the (R_C, R_B) space confirms three-regime consistency (Supplementary Table 7).

Table 2 Regime-dependent improvements across model families relative to a PyTorch default baseline. Platform: NVIDIA A100 (80 GB), batch size = 8, sequence length = 2048. Operating regime: coordination-dominated ($R_C \geq 0.5$, $R_B \geq 0.4$). Values: mean ± 95 % bootstrap CI over $n_{\text{eff}} = 150$ measurement windows.

Model	Task	Load time reduction	Latency reduction	Swap reduction
Llama-3 8B	Generation	$52.0 \pm 1.8\%$	$22.0 \pm 1.2\%$	$23.0 \pm 1.5\%$
GPT-J 6B	Summarise	$55.3 \pm 2.1\%$	$25.0 \pm 1.4\%$	$25.1 \pm 1.7\%$
Llama-4 17B	Generation	$57.4 \pm 2.3\%$	$30.8 \pm 1.6\%$	$26.8 \pm 1.8\%$
Mixtral 8×7B	MoE	$59.1 \pm 2.5\%$	$32.5 \pm 1.9\%$	$27.5 \pm 2.0\%$
ViT-H/14	Vision	$51.2 \pm 1.7\%$	$18.2 \pm 1.1\%$	$20.9 \pm 1.4\%$

Scientific computing workloads.

The regime structure extends to HPC-style scientific inference (Table 3, lower rows). In ESMFold protein structure prediction [19], long-sequence batches (length 1024, batch 8) accumulate MSA embeddings that shift R_C from 0.67 (coordination-dominated) to 0.43 (capacity-limited) during inference; Orion detects the transition and recovers $19.3 \pm 1.6\%$ latency, while FlexGen's aggressive offloading degrades by +6.4% once capacity saturates. In Pangu-Weather climate downscaling [20], large spatial tensor transfers hold the system in the I/O-limited regime ($R_C = 0.54$, $R_B = 0.26$); swap-reduction recovers $13.8 \pm 1.3\%$ latency, whereas FlexGen amplifies transfer pressure by +5.1%. In both cases the (R_C, R_B) classifier correctly identifies the regime, confirming that the two-ratio framework applies to scientific computing workloads with large spatial or sequence tensors.

Isolating the classifier's contribution.

Comparing Orion$_{\text{HW}}$ (hardware subsystems only, classifier disabled, strategy fixed) against Orion$_{\text{Full}}$ (classifier running every 100 ms) reveals that hardware optimisations alone yield −3.8 to −8.3% latency reduction; the regime classifier contributes an *additional* −1.1 to −18.8% (21–69% of total gain on LLM workloads), with the largest contribution in the coordination-dominated regime (69%) where strategy choice has the most leverage. For workloads with mid-inference transitions (BLIP2, RAG), the classifier contributes 66–76% of total gain (Supplementary Table 10).

Table 4 shows that energy efficiency is also regime-dependent, consistent with cluster-level evidence that the energy profile of LLM serving depends strongly on operating conditions [21]. To quantify the regime effect on energy, we re-ran the same Llama-3 8B workload at fixed batch size = 8 and sequence length = 2048 while sweeping R_B from 0.75 (coordination-dominated) down to 0.15 (I/O-limited), holding all other variables constant. In I/O-limited configurations ($R_B < 0.20$), throughput-per-Watt falls by 3.2× compared with coordination-aligned operation ($R_B \geq 0.40$); using the IEA (2024) global average carbon intensity of 494 gCO_2/kWh [4], this corresponds to a rise from 15.8 g to 42.5 g of CO_2 per 10^6 tokens generated. This indicates that regime misalignment—not hardware limitations alone—is a root cause of energy inefficiency.

Table 3 Regime structure and latency change across AI and scientific computing workloads under all tested orchestration strategies. All experiments: NVIDIA A100 (80 GB), batch size = 8. Regime assigned by the runtime classifier (93.4% accuracy). Values: mean ± 95 % bootstrap CI over $n_{\mathrm{eff}} = 150$ measurement windows; latency change relative to PyTorch default. **Bold**: best performer per workload. *Italicised*: latency increase (worse than baseline). *vLLM not applicable to workloads without an autoregressive KV cache.

Workload	Observed regime	R_C	R_B	FlexGen	vLLM	Orion (adaptive)
BLIP2 VQA	Coord.→I/O	0.53 → 0.51	0.43 → 0.31	*+1.8 ± 0.7%*	−3.2 ± 0.6%	**−11.2 ± 1.4%**
RAG (FAISS)	Cap.→Coord.	0.46 → 0.58	0.42	−5.1 ± 0.8%	−8.3 ± 1.1%	**−17.4 ± 1.8%**
YOLOv8	I/O-limited	0.49	0.22	*+7.8 ± 0.9%*	n/a*	**−9.5 ± 1.2%**
ESMFold [19]	Coord.→Cap.	0.67 → 0.43	0.51	*+6.4 ± 0.9%*	−5.2 ± 0.7%	**−19.3 ± 1.6%**
Pangu-Weather [20]	I/O-limited	0.54	0.26	*+5.1 ± 0.9%*	n/a*	**−13.8 ± 1.3%**

Table 4 Energy efficiency (throughput per Watt, normalised to PyTorch = 100). Platform: NVIDIA A100 (80 GB); model: Llama-3 8B; batch = 8; sequence length = 2048. Power measured via NVML at 100 ms intervals; coordination-dominated regime ($R_C \geq 0.5$, $R_B \geq 0.4$). Values: mean ± 95 % bootstrap CI over $n_{\mathrm{eff}} = 150$ windows. CO_2 equivalents use the IEA (2024) global average carbon intensity of 494 gCO_2/kWh [4]; grid-mix variability dominates CO_2 uncertainty (±30 %).

Method	Avg. power (W)	Throughput (tok/s)	Tput/Watt (norm.)
PyTorch (default)	310 ± 4	6,200 ± 80	100.0
FlexGen [5]	315 ± 5	6,800 ± 90	108.0 ± 2.1
SwapAdvisor [7]	320 ± 5	7,040 ± 95	110.0 ± 2.2
vLLM [8]	318 ± 5	7,120 ± 90	112.0 ± 2.1
Orion	322 ± 5	7,620 ± 100	118.5 ± 2.4

2 Discussion

The regime-dependent framework establishes a practically important negative result: beyond regime boundaries, additional orchestration can predictably *increase* latency and energy expenditure by exposing synchronisation overhead without improving throughput. In capacity-limited regimes, the absence of effective locality imposes unavoidable memory-access costs that coordination cannot offset. In I/O-limited regimes, physical bandwidth constraints dominate once cross-tier transfer approaches sustained limits; any additional coordination overhead is additive rather than compensatory. This is analogous to, but mathematically distinct from, thermodynamic phase transitions. In a thermodynamic system of diverging size, the order parameter changes discontinuously and the susceptibility (analogous to S) diverges at the critical point. Here the "system size" is finite: the number of transformer layers L is fixed, and both S and the boundary width carry $O(1/L)$ corrections from the continuum limit. The transitions are therefore more precisely *finite-size crossovers*—abrupt but bounded changes in the dominant latency term at a well-defined critical ratio. Empirically, $S = 4.12$ at θ_C ($> 2 \times S^*$) demonstrates that the crossover is operationally indistinguishable from a true transition at any practically relevant model scale, even though S remains finite. This precision resolves the "phase-like" qualifier used throughout the main text: the transitions are genuine crossovers whose sharpness scales with L, and the analogy to physical phase transitions motivates the dimensionless-ratio formulation and may extend the framework to other hierarchical-memory systems beyond AI inference.

A regime-based lens provides a unifying explanation for heterogeneous—and sometimes contradictory—results in prior systems work. Offloading, caching, and swapping strategies that implicitly assume operation within coordination-dominated regimes deliver substantial gains when that assumption holds. When deployed in configurations where capacity or bandwidth constraints dominate, their effectiveness diminishes sharply. Differences in fast-memory capacity, sustained transfer bandwidth, and workload characteristics can place nominally similar systems in different regimes, producing divergent benchmark outcomes without obvious explanation. Making the operating regime explicit therefore directly improves the interpretability and generalisability of inference systems research.

A natural counterargument is that emerging memory technologies—HBM capacity expansion, CXL-based memory pooling, disaggregated memory—will render regime boundaries irrelevant. Our analysis shows these technologies can *shift* boundaries by increasing intermediate capacity or improving sustained bandwidth, but they do not remove the underlying phase structure. Once transfer demand approaches a tier's sustained bandwidth ceiling, or once working-set residency fails in fast memory, the dominant term in the latency decomposition changes discontinuously regardless of absolute capacity or bandwidth values. More sophisticated software strategies—aggressive prefetching, deeper computation–transfer overlap—can redistribute visible latency within coordination-dominated regimes but cannot violate physical capacity and sustained-bandwidth limits. Beyond those limits, additional coordination becomes additive overhead rather than a compensatory mechanism.

The framework directly motivates adaptive inference systems. Rather than applying a fixed orchestration strategy uniformly, a regime-aware controller can sample the latency decomposition (T_{mem}, T_{swap}, T_{sync}) at low overhead and classify the operating regime—a task our depth-3 decision-tree classifier accomplishes with 93.4% accuracy across diverse workloads at $<$0.1 ms cost per classification. The first question for any memory-bound inference system should be not "how much more coordination?" but "in which regime does the system operate?"

The regime framework implies concrete prescriptions for each operating condition. In the *capacity-limited* regime ($R_C < 0.50$), the primary lever is fast-memory residency: increase HBM capacity, reduce batch size, or apply weight quantisation or one-shot pruning [22] to shrink W; additional coordination overhead has negligible return ($\leq$ 5% in our experiments, Table 1). In the *coordination-dominated* regime ($R_C \geq 0.50$, $R_B \geq 0.40$), pipelining, KV-cache management, and cross-tier overlap all provide measurable gains (18–27%); this is the regime for which existing orchestration tools are effectively designed. In the *I/O-limited* regime ($R_B < 0.40$), the correct intervention is to reduce compulsory transfer volume—via smaller swap granularity, selective layer offloading, or on-device quantisation—rather than adding scheduling complexity; aggressive offloading strategies (FlexGen, DeepSpeed-Inference) can *worsen* latency by 8–12% in this regime by amplifying an already saturated transfer path. These prescriptions are immediately actionable from two ratios computable before deployment. Crucially, both thresholds are predictable a priori without exhaustive sweeps: $\theta_C = 0.50$ follows analytically from the majority-eviction condition (analytically derived, not dependent on specific hardware constants; consistent within ± 0.02 across five platforms, mean error $\leq 2.4\%$), and θ_B is estimable from three measurements—sustained bandwidth B_{slow}, single-run computation time T_{comp}, and DMA log-derived amplification factors—obtainable in a 10-minute profiling session (mean error $\leq 4.7\%$ on PCIe-based platforms; Supplementary Table 4).

The framework applies to any workload whose active working set cannot fully reside in fast memory, including the scientific computing cases validated above, and provides a principled foundation for regime-aware design across increasingly complex memory hierarchies.

Several limitations motivate future work. The classifier's 93.4% accuracy is in-distribution: training and evaluation used the same five platforms; generalisation to unseen hardware (e.g., CXL-pooled memory) is an open direction. Multi-node clusters shift θ_B (4-node experiments: $\Delta\theta_B \approx +0.05$), but regime fragmentation at scale is uncharacterised. Hysteresis when workloads traverse boundaries in both directions suggests nonlinear cache-reuse effects exploitable in adaptive scheduling. For mixture-of-experts models, $W_{\text{eff}} = W_{\text{param}} \times (k/N_{\text{exp}}) + W_{\text{act}} + W_{\text{kv}}$ shifts R_C relative to dense activation; speculative decoding may place draft and target models in different regimes simultaneously, requiring per-model classification. Optimal within-regime orchestration, particularly in the coordination-dominated case, remains open; combining boundary theory with reinforcement learning or Bayesian optimisation is a promising direction.

Methods

Experimental framework: Orion

Orion is a lightweight experimental framework for coordinating accelerator memory, host DRAM, and NVMe storage as a unified hierarchy during AI inference. It is used here as a *measurement instrument* rather than a production serving engine: its role is to enable controlled traversal of the regime space by making the latency components T_{comp}, T_{mem}, T_{swap}, and T_{sync} individually observable at runtime.

Orion comprises three instrumented subsystems, borrowing the OS notion of memory ballooning [23] to manage the whole VRAM–DRAM–NVMe hierarchy together.

(1) **GPU Process Allocator**: In multi-model serving, GPU context loading conflicts and VRAM interference dominate. The GPU Process Allocator reserves GPU execution resources up front—context IDs, memory pools, and kernel queues are pre-assigned—keeps memory segments separated across models to limit cache pollution, and co-schedules I/O-intensive and compute-intensive models so the GPU does not

idle. It runs three policies: pre-defined execution slot reservation, context separation with fixed memory pools, and priority-based scheduling between I/O-bound and compute-bound models.

(2) **Task Memory Manager**: This module pairs task-level memory control with NUMA-aware policies [24]: aligning processes with the right memory nodes via `mbind()` and `numa_alloc_onnode()` reduces remote-access delay, and a `jemalloc`-based custom heap separates small objects from large tensors to keep fragmentation manageable. Memory access latency is modelled as $T_{\text{access}} = T_{\text{local}} + \gamma \cdot T_{\text{remote}}$, where γ is the penalty factor for a remote NUMA access. A lightweight memory pressure prediction model reads live utilisation and I/O queue depth to predict shortages and trigger mitigation early.

(3) **GPU Booster with Swap Support**: This module walks the execution graph to spot inactive tensors and overlaps their movement with computation via direct PCIe DMA and asynchronous I/O, aligning swap units to 4 MB boundaries to avoid wasted transfers. Selective tensor offloading uses execution-graph analysis to flag tensors not immediately needed—inactive attention KV caches, intermediate checkpoints—as swap candidates. Priority-aware, block-aligned swapping weighs both tensor size and reuse frequency; on a cache miss, partial reloads keep transfer volume small. With these mechanisms, Orion served models up to 17B parameters on GPUs with less than 80 GB of VRAM without observed instability.

Regime probing methodology

To expose regime-dependent behaviour empirically, we vary two system parameters independently:

(i) Capacity ratio $R_C = C_{\text{fast}}/W$: the fraction of model weights pinned in GPU HBM versus spilled to host DRAM is adjusted programmatically via Orion's tensor-placement API. At each target R_C value, weights are partitioned layer-by-layer using `cudaMemcpy` to host, leaving a controlled C_{fast} of HBM resident.

(ii) Transfer-pressure ratio $R_B = B_{\text{slow}} \cdot \Delta t/D$: host-to-GPU PCIe bandwidth is throttled using a software token-bucket rate-limiter integrated into Orion's DMA engine. The limiter wraps each `cuMemcpyHtoDAsync` call with a counting semaphore that caps outstanding transfer bytes per 10 ms window. Target bandwidth levels are validated against NVIDIA NVML `nvmlDeviceGetPcieThroughput` readings; achieved bandwidth deviates from target by $\leq 3\%$. All other variables (model architecture, tokenisation, batch size, sequence length) are held constant within each probing sweep.

Cross-validation of circularity. Because Orion's software token-bucket controls R_B on the same system where S is measured, we explicitly address the circularity concern by cross-validating sharpness coefficients on two platforms where sustained bandwidth is hardware-limited and the software throttle is *inactive*: AMD MI250 (HBM2e bus, peak 3.2 TB/s) and Intel Optane-PMem (hardware-governed PMem bandwidth ceiling). On both platforms, $S > 2.0$ at θ_B is confirmed without any software bandwidth manipulation: $S = 3.61 \pm 0.29$ on MI250 and $S = 3.88 \pm 0.33$ on Optane-PMem (bootstrap 95 % CI, $n_{\text{sweep}} = 5$). These measurements establish that the sharpness of the I/O-limited transition is a property of the physical bandwidth constraint, not an artefact of Orion's instrumentation.

Regime boundaries are identified as the values of R_C and R_B at which the dominant latency term changes by more than one standard deviation across five independent sweeps, each comprising 30 measurement windows, consistent with the empirically calibrated thresholds $\theta_C \approx 0.50$ and $\theta_B \approx 0.40$ (see Supplementary Note 1).

Scientific computing workload details

ESMFold protein structure prediction experiments used the production 690 M-parameter evoformer model (ESMFold v1.0 weights, Meta AI) under realistic inference conditions: variable-length sequence batches (lengths 512–1024, batch size 8) on NVIDIA A100 (80 GB), matching the memory-pressure profile of production structural biology pipelines. Pangu-Weather climate downscaling experiments used the full 3D Earth System Model (64.5 M parameters, production weights from Bi et al. [20]) with full-resolution ERA5 input tensors (1440 × 721 horizontal grid, 13 pressure levels), representing the spatial tensor volumes encountered in operational medium-range forecasting. Both workloads ran on the same NVIDIA A100 (80 GB) hardware environment as all other experiments; no surrogate models or reduced-scale proxies were used. W and D for each workload were measured via ORION's allocator-level tracker under the batch configurations described above; R_C and R_B values reported in Table 3 are step-averaged over 30 contiguous 10 s windows per sweep.

Baselines and workloads

Orchestration baselines span four categories: CPU offloading (FlexGen [5], DeepSpeed-Inference [6]), GPU–CPU swapping (SwapAdvisor [7]), KV-cache management (vLLM [8]), and a PyTorch default serving baseline. Language workloads use Llama-3 8B, GPT-J 6B, Llama-4 17B, and Mixtral 8×7B; the vision workload uses ViT-H/14.

Hardware platforms

Primary experiments use two NVIDIA A100 (80 GB HBM) GPUs, 512 GB DDR5 host DRAM, and an NVMe Gen4 SSD (software stack: PyTorch 2.4, CUDA 12.2, vLLM 0.5.3). Cross-platform validation uses *real hardware* on four additional platforms: Google TPU v4 pods (JAX 0.4.26), AWS Inferentia2 endpoints (NeuronSDK 2.18), AMD MI250 GPUs (ROCm 6.0), and Intel Xeon + Optane-PMem servers. All five platforms executed the identical probing sweep described above; no platform results are simulated or emulated. Reported transition points vary by ≤ 0.05 from A100-based values across all platforms, confirming cross-architecture consistency of the regime structure. Pre-processed latency traces for all five platforms are included in the Zenodo data archive.

Operational definition of working-set size W

The active working-set size per inference step is defined as $W = W_{\text{param}} + W_{\text{act}} + W_{\text{kv}}$, where W_{param} is the parameter byte footprint, W_{act} is the peak activation memory over a prefill+decode cycle ($B \cdot \ell \cdot L \cdot 4d_{\text{model}} \cdot s_{\text{dtype}}$), and W_{kv} is the KV-cache footprint ($2BLHd_h\ell\, s_{\text{dtype}}$). FlashAttention-style fused kernels [25] reduce W_{act} but affect neither W_{param} nor W_{kv}. W is measured at runtime by ORION's allocator-level memory tracker, which records the high-water mark of GPU-resident tensor bytes per step. Analytical values and per-model breakdowns are given in Supplementary Table 9.

Measurement protocol

Replication structure. Each experimental condition consists of $n_{\text{sweep}} = 5$ independent probing sweeps. Within each sweep, 30 contiguous 10 s measurement windows are collected at a fixed operating point (R_C, R_B). Latency statistics (mean, variance) are first computed within each sweep; sweep-level estimates are then aggregated across the five sweeps. Bootstrap resampling (1,000 replicates, sampling windows within sweeps with replacement) yields 95 % confidence intervals; these are reported in the main text and Supplementary Note 2. The effective sample size is $n_{\text{eff}} \approx 150$ per condition.

Sharpness coefficient S and numerical differentiation. The sharpness coefficient $S = |d \ln T_{\text{total}}/d \ln R|$ is estimated at each operating point via symmetric finite differences over adjacent R grid steps ($\Delta R = 0.05$):

$$S(R_i) = \left| \frac{\ln T(R_{i+1}) - \ln T(R_{i-1})}{\ln R_{i+1} - \ln R_{i-1}} \right|,$$

where $T(R_j)$ is the sweep-mean latency at grid point j. No additional smoothing is applied; the grid spacing $\Delta R = 0.05$ is fine enough to resolve the ± 0.05 boundary width identified empirically. Reported S values (e.g. $S = 4.12 \pm 0.31$ near θ_C) are means and bootstrap 95 % CIs over the five sweep-level estimates of $S(R_i)$.

Statistical significance. For the primary claim that $S > S^* = 2.0$ near regime boundaries, we apply a pre-specified one-sided Wilcoxon signed-rank test (null: $S \leq 2.0$; alternative: $S > 2.0$) to the five sweep-level S estimates. The one-sided framing is pre-specified: the directional alternative follows directly from the a priori definition of abruptness—$S^* = 2.0$ was fixed before data collection, corresponding to a 10% change in R producing >20% latency change, a threshold chosen on physical grounds independent of the data. With $n_{\text{sweep}} = 5$, the minimum achievable one-sided p-value under the Wilcoxon signed-rank distribution is $p = 0.031$ (exact; achieved when all five sweep estimates exceed S^*). All reported near-boundary conditions meet this criterion; interior conditions yield $p > 0.40$, consistent with H_0. *Statistical power.* Because the observed effect is large relative to its dispersion ($S = 4.12 \pm 0.31$ at θ_C, i.e. the boundary $S^* = 2.0$ lies $\approx$6.8 sweep-level standard deviations below the mean), every sweep estimate exceeds S^* with near-certainty; a Monte Carlo power analysis (10,000 draws from the empirical sweep distribution) gives power > 0.99 at $n_{\text{sweep}} = 5$ for the observed effect size. The test is therefore not underpowered for effects of this magnitude—the small n_{sweep} limits the *minimum reportable* p-value (0.031), not the ability to detect the transition.

Warm-up and steady-state. Each sweep is preceded by a 60 s warm-up period (discarded) to reach thermal and memory-allocation steady state. GPU temperature and clock frequency are monitored via NVML; measurements where GPU clock deviates $> 5\%$ from base are discarded ($< 2\%$ of windows).

T_{comp} is isolated via asynchronous CUDA events inserted immediately before and after each matrix-multiplication kernel batch; event synchronisation is performed off the critical path to avoid serialisation overhead.

T_{mem} is derived from NVIDIA Performance Counters (CUPTI) sampling HBM read bandwidth and L2 cache-miss rate at per-kernel granularity; the memory-access latency is computed as miss_volume / effective_HBM_bandwidth. The HBM miss-penalty per byte ρ (platform constant in Eq. (3)) is measured from the same CUPTI counters during a synthetic streaming kernel that issues controlled cache-miss loads; ρ ranges from 2.1 to 6.2 ns/byte across the five platforms and platform-specific values are listed in Supplementary Table 1.

T_{swap} is measured directly from DMA transfer logs produced by ORION's asynchronous transfer engine, which records start and end timestamps for every host$\leftrightarrow$device copy.

T_{sync} is measured *directly* via explicit synchronisation-barrier profiling rather than as a residual. Specifically, CUDA barrier events (`cudaStreamSynchronize` and `cudaDeviceSynchronize`) are wrapped with high-resolution `clock_gettime(CLOCK_MONOTONIC)` calls; NCCL collective operations are instrumented at the `ncclAllReduce` boundary using CUPTI callback APIs. The sum of all barrier and collective durations is reported as T_{sync}. To validate this measurement, we verify that the four-term sum $\hat{T}_{\text{total}} = T_{\text{comp}} + T_{\text{mem}} + T_{\text{swap}} + T_{\text{sync}}$ agrees with wall-clock latency within $\leq 3\%$ across all configurations; the residual $|T_{\text{wall}} - \hat{T}_{\text{total}}|$ averages 1.8% (s.d. 0.6%), confirming measurement completeness.

Energy consumption is measured using NVML power sampling at 100 ms intervals and integrated over each inference request.

Ablation methodology

To confirm that observed regime-dependent behaviour is not driven by any single ORION subsystem, we perform a component ablation by disabling each subsystem independently (process allocation, NUMA binding, GPU-tailored swapping) and repeating the full regime probing sweep. Regime boundaries shift by ≤ 0.03 across ablation variants, confirming that the abrupt regime transitions are intrinsic to the hierarchical memory system rather than artefacts of ORION's implementation.

Regime classifier

A depth-3 CART decision tree is trained on runtime statistics (swap-to-computation ratio, cache hit rate, DMA utilisation) collected from 2,400 probing measurements covering all three regimes and five hardware platforms. The classifier is evaluated via stratified five-fold cross-validation; mean accuracy is 93.4% (s.d. 1.2%). Inference cost is <0.1 ms per classification step, negligible relative to request latency. **Important caveat:** the 93.4% accuracy figure is an *in-distribution* result—training and evaluation use measurements from the same five hardware platforms. Generalisation to unseen hardware configurations, novel memory-tier arrangements, or qualitatively different bandwidth profiles has not been validated and is an open direction for future work.

Data availability

Pre-processed latency traces for all reported experiments—including the NVIDIA A100, Google TPU v4, AWS Inferentia2, AMD MI250, and Intel Optane-PMem platforms—together with the regime-classifier training data are available at https://github.com/leemgs/orion. Raw measurement logs (`.jsonl` format, one entry per 10 s measurement window) will be archived with a DOI on Zenodo upon acceptance; they are available from the corresponding author on reasonable request during review.

Code availability

The Orion measurement framework—including the DMA rate-limiter, NUMA-aware memory manager, CUPTI instrumentation wrappers, and the regime probing scripts used to generate all figures and tables—is openly available at https://github.com/leemgs/orion, with installation and build instructions for all five hardware platforms. The `src/experiments/` directory additionally provides simulation-based reproduction scripts (a parameterised `SimulatedBackend`) that reproduce the qualitative regime structure without physical access to the five platforms; all quantitative values reported in this paper derive from the real-hardware measurements described in Methods.

Author Contributions

G.L.: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Competing Interests

The author declares no competing interests.

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